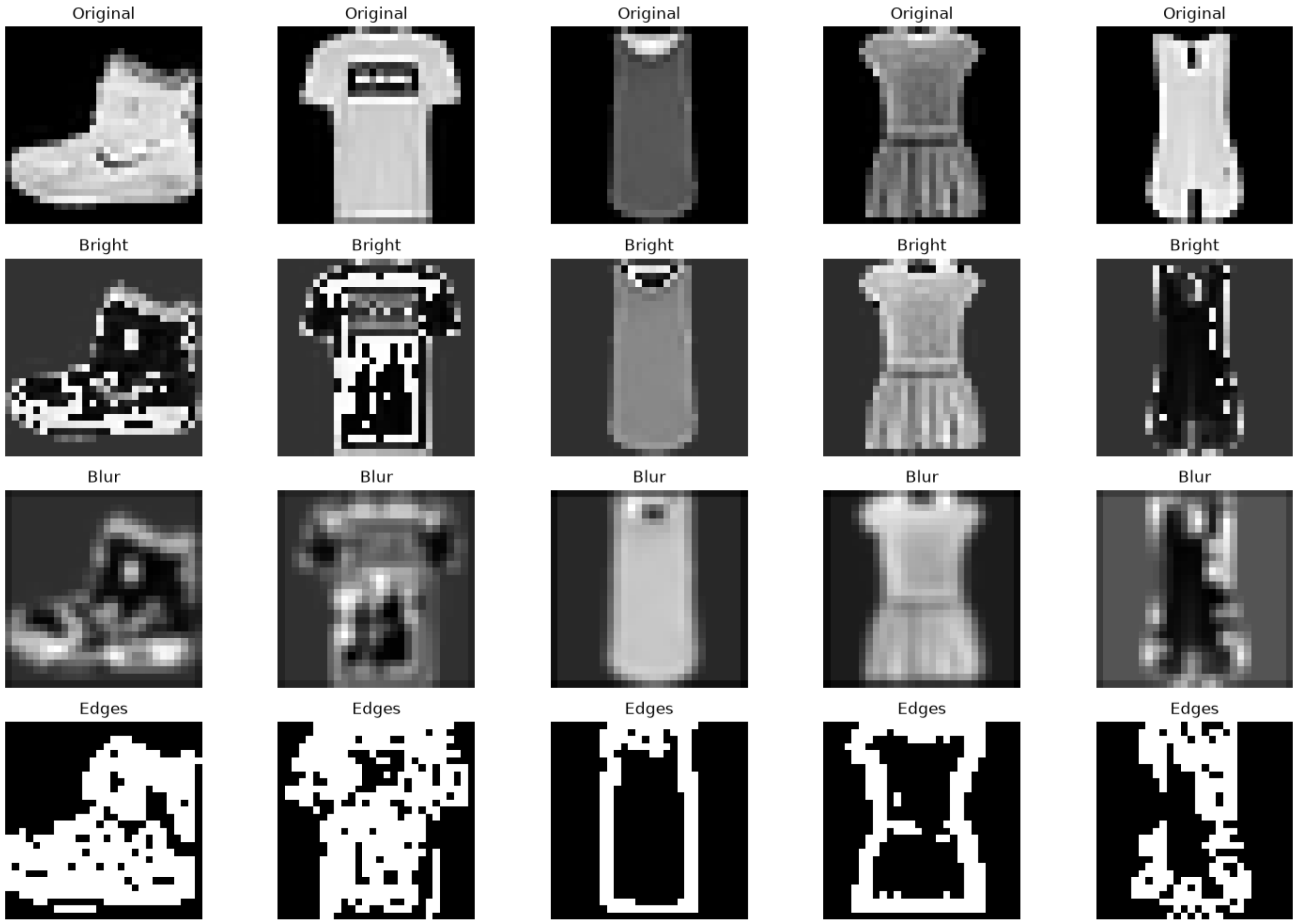
In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.datasets import fashion_mnist

(fashion_x, _), _ = fashion_mnist.load_data()

pixel = 100
new_pixel = min(pixel + 50, 255)

print("Original Pixel:", pixel)
print("After Brightness +50:", new_pixel)

bright_images = []
blur_images = []
edge_images = []

for i in range(5):
    img = fashion_x[i]

    bright = np.clip(img + 50, 0, 255)

    blur = tf.nn.conv2d(
        bright.reshape(1, 28, 28, 1).astype(np.float32),
        tf.constant([[[[1]], [[2]], [[1]]],
                     [[[2]], [[4]], [[2]]],
                     [[[1]], [[2]], [[1]]]], dtype=tf.float32) / 16,
        strides=1,
        padding='SAME'
    ).numpy().squeeze()

    gx = np.abs(np.diff(blur, axis=1))
    gy = np.abs(np.diff(blur, axis=0))

    gx = np.pad(gx, ((0,0),(0,1)))
    gy = np.pad(gy, ((0,1),(0,0)))

    edges = np.sqrt(gx**2 + gy**2)
    edges = (edges > 20) * 255

    bright_images.append(bright)
    blur_images.append(blur)
    edge_images.append(edges)

plt.figure(figsize=(15,10))

for i in range(5):
    plt.subplot(4,5,i+1)
    plt.imshow(fashion_x[i], cmap='gray')
    plt.title("Original")
    plt.axis('off')

for i in range(5):
    plt.subplot(4,5,i+6)
    plt.imshow(bright_images[i], cmap='gray')
    plt.title("Bright")
    plt.axis('off')
```

```
for i in range(5):
    plt.subplot(4,5,i+11)
    plt.imshow(blur_images[i], cmap='gray')
    plt.title("Blur")
    plt.axis('off')

for i in range(5):
    plt.subplot(4,5,i+16)
    plt.imshow(edge_images[i], cmap='gray')
    plt.title("Edges")
    plt.axis('off')

plt.tight_layout()
plt.show()
```

```
Original Pixel: 100
After Brightness +50: 150
```



In [ ]: